

Data Analysis Plan – Improving Diabetes Management with Community Education

Introduction

The purpose of this Data Analysis Plan is to summarize the proposed study, as well as provide reasoning behind the recommended statistical methods used to gain insight on the effectiveness of education towards diabetes self-care.

Aim of Study

The aim of this study is to test the effectiveness of a community education program in improving diabetes self-care. The desired outcome is to help the participants in this study reduce their risk of Type 2 Diabetes with education, which can be measured in their A1C%, Weight Loss, and Knowledge of Diabetes self-care. This Data Analysis Plan is provided to help guide the researchers with the statistical aspects of the study.

Research Question

For those in a Weight Loss Group or at Risk of T2D, does a community educational outreach program improve diabetes management after 8 weekly meetings?

Study Design

This study will be performed at the Minneapolis Health Clinic over an 8-week period during the summer, with the population being current patients at the clinic. The sample of participants in this study consists of those who are at risk of T2D or are in a Weight Loss Group recommended by their physician. The inclusion criteria for participation are limited to those at risk of T2D, overweight, age 18-69, and willing to attend a weekly group class. The study design is a mixed method with most statistical tests being Single Group Pretest-Posttest. There are typically 10 participants for this study, with this current run expecting 6 to 10 participants.

This single group of participants will take a Pre-survey as a baseline to compare with a post-survey after the intervention is completed. The Pre-survey will ask participants for their:

- Most current A1C% within the last 90 days, numeric
- Current weight, numeric
- Self-stated knowledge, Likert Scale.
 - Suggested levels: No understanding, Some understanding, So-so understanding, Moderate understanding, Expert understanding.

Weight will also be recorded at each meeting as well. All-in-all, it's unnecessary to statistically consider the participants' weight change over the weeks; a simple before and after is sufficient. If it is desired to use the weekly data, the results would show, generally, which weeks weight loss was high, or low. This doesn't seem to line up with the expected results being long-term lifestyle improvement.

It's our recommendation to forgo the self-stated knowledge question, in favor of a brief knowledge assessment. This could be used to reduce confounding due to participants falsely claiming higher comprehension. I recommend including enough questions to cover each topic (Weight Loss, Diet Choices, Increasing Activity, Stress Management Skills, Long-term Lifestyle Change Commitment, etc.), but not so many questions that it's exhausting to complete. This should be converted into a cumulative point score, similar to Prediabetes Risk Test.

The Post survey will update these variables of interest for a simple before and after comparison, with A1C% measured 3 months after the start of the intervention to capture the changes.

There is only one independent variable, which is the participant ID, a nominal data type. It is necessary to assign each participant an ID to pair the data together. At first a name can be used, but to ensure participant anonymity, an ID is highly recommended for data processing and statistical analyses.

The general workflow of the study should go as follows:

1. Develop an Analysis plan (This document), and lesson plan.
2. Spread the word/recruit participants for the study.
3. First day with the participants: orientation for the participants, conduct the Pre-survey and Prediabetes Risk Test, gather A1C% if applicable, perform first lecture and Q/A if time allows.
4. Subsequent weeks with participants: Measure participant weight, and perform lecture and Q/A.
5. Final week with participants: Perform final lecture and Q/A, conduct Post survey.
6. Three months after first meeting: Follow up with participants, gather A1C%.
7. Perform data entry and cleaning.
8. Perform Statistical Analyses

Statistical Methods

Our primary goal is to test for differences in the variables of interest before the intervention with after. The first step involves getting a better understanding of the data, so we must perform exploratory data analysis to get a deeper understanding of it all. This includes calculating descriptive statistics as a whole and between groups. Creating graphs, such as line graphs to emphasize the change over time, and histograms to check the distribution of the data.

When it comes to statistical tests, there is some finesse when it comes to selecting the best statistical method for each one. Typically, there are tests to compare means only if their assumptions are met, and an alternative if there are violated assumptions: a nonparametric equivalent which compares medians. If there are violated assumptions, and a nonparametric equivalent is not used, there will likely be false negatives or false positives.

For assumption of normality, we want to ensure that the difference of the pairs is normally distributed with constant variance. A Shapiro-Wilks test can test for normality. The data can also be transformed to encourage normality, in the case of weight change. When it comes to checking for outliers, a boxplot can help determine if there are any. Each of these assumption and statistical tests can be performed in a variety of statistical software packages, such as R.

Change in A1C%

For A1C%, if the differences between pairs of data are approximately normally distributed, independent, constant difference between groups, and free of extreme outliers, comparing the difference of the pairs can be done using a One-Sided Paired t-Test with the null hypothesis $\mu_d = 0$ and the alternative hypothesis being $\mu_d > 0$, where μ_d is the difference of pairs of A1C%. Otherwise, a One-sided Wilcoxon Signed Rank Test is the nonparametric equivalent that compares the medians instead. A statistically significant test will indicate that the mean A1C% has decreased after the intervention.

To graphically present this: for the Paired t-Test, I would recommend a bar chart or line graph measuring the mean A1C% with the confidence intervals for both before and after; and for the nonparametric test, Box plots of the A1C% for both the before and after on the same scale should suffice.

Change in Knowledge

To test the change in knowledge, whether the researchers proceed with the knowledge assessment or Self-stated knowledge, the approach should be the same. Use a One-sided Paired t-Test with the null hypothesis being $\mu_d = 0$, and alternative hypothesis: $\mu_d > 0$, where μ_d is the mean difference in Knowledge. Use the Shapiro-wilks test to check normality of differences between pairs, and box plots of the differences to check for outliers. If the data violates the assumptions, then the One-sided Wilcoxon Signed Rank Test is the nonparametric equivalent with the same alternative hypothesis that compares

medians instead. A statistically significant result will indicate that the mean knowledge has increased after the intervention.

For graphical presentation, the same as A1C% should work.

Change in Weight

For weight change, as the participants are expected to lose 5-7% of their body over the 6 months period mentioned in Prevent T2 handouts, pairs of weight will very likely be non-normally distributed and not have constant variance between before and after, as a higher start weight will have more variability. To fix this issue it's recommended to test the % weight change.

$$\% \text{ Weight Change} = \frac{\text{Weight}_{\text{after}} - \text{Weight}_{\text{before}}}{\text{Weight}_{\text{before}}} \cdot 100\%$$

Example of Mean Percent Change in Weight			
ID	Weight Before	Weight After	% Weight Change
1	160	155	$\frac{155 - 160}{160} = -3.1\%$

After the transformation, the assumptions should be assessed for approximate normality and outliers. Use Shapiro-Wilks test for normality, box plots to determine outliers again. If the assumptions are held, then a One sample t-Test comparing the mean percent weight change with an alternative hypothesis of $\mu < 0$ will test if the weight after is significantly less than the weight before. If the assumptions are not held, then a Wilcoxon Signed Rank Test can be used instead which will compare the ranks will work.

To graphically present this, plotting the mean percent weight change with confidence intervals against the $y = 0$ axis could work.

Power

Power, in statistics, is the probability of rejecting a null hypothesis, given that the null hypothesis is false. This is largely dependent on sample size, significance level, the difference in assumed and actual means, and the assumed or actual standard deviation.

For example, the power for the t-Test for weight change for an estimated 2% weight loss, assumed standard deviation of 2% over the two-month study with a sample size of 7, would have a power of 0.7. This is rather low, compared to a generally accepted target of 0.8 to 0.9, and indicates a moderate chance for the test to have a false negative result, and limits the generalizability of the test. The power can be increased by having a larger sample size, if the actual standard deviation is smaller than the assumed, or if there's a bigger difference to detect. Here is a table of example power scenarios with assumed standard deviation of 2%:

Sample Size required for specific power level at % Weight Change			
Power	Alternative Percent Weight Change		
	1.5%	2%	2.5%
.7	10	7	5
.8	13	8	6
.9	17	11	8